1. Aim of the practical: To understand and demonstrate the usage of grep, e-grep, and f-grep **commands in Linux for searching and filtering text patterns efficiently within files or streams.**

2. Objective:

- Learn how to use the basic **GREP** command to search for patterns in files using regular expressions.
- Explore the **EGREP** (extended GREP) command to work with extended regular expressions that allow additional metacharacters for pattern matching.
- Learn how to use the **FGREP** command to search for fixed strings (without regular expressions) in files.
- Understand the performance implications and best practices for choosing between these commands depending on the use case.

3. Hardware and Software Requirements:

a) **Disk Drive**: Physical disk (HDD/SSD) with enough space.
Computer System: Host system running virtualization software.
b) **Operating System**: OS on the virtual machine (e.g., Fedora).

4. Commands/Steps:

- Touch and cat command to create and access file

```
nikhilyadav@fedora:~$ touch story.txt
nikhilyadav@fedora:~$ cat > story.txt
there ia hidden treasure in a forest.
only those with sharp eyes will find it.
but beware, it's by giant wolves.
!
```

Grep commands

- **-- color: By default, this option enables colorized output, highlighting the matching patterns.**

```
nikhilyadav@fedora:~$ grep  "wolves" story.txt
but beware, it's by giant wolves.
```

- Grep -r “string” * command: searching for a string recursively in all directories

```
nikhilyadav@fedora:~$ cat >jungle.txt
My name is Nikhil.
Nikhil is in mca .
^C
nikhilyadav@fedora:~$ grep -r "Nikhil"
grep: .cache/tracker3/files/http%3A%2F%2Ftracker.api.gnome.org%2Fontology%2Fv3%2Ftracker%23D
uments.db-wal: binary file matches
grep: demo.txt: Permission denied
jungle.txt:My name is Nikhil.
jungle.txt:Nikhil is in mca .
```

- Grep -I “string” command is use to ignore case sensitive

```
nikhilyadav@fedora:~$ cat > jungle.txt
comands are fun... comands are used to direct some task.. like grep comands...like f-grep comands
^C
nikhilyadav@fedora:~$ grep "comands" jungle.txt
comands are fun... comands are used to direct some task.. like grep comands...like f-grep comands
nikhilyadav@fedora:~$ grep -i "comands" jungle.txt
comands are fun... comands are used to direct some task.. like grep comands...like f-grep comands
nikhilyadav@fedora:~$
```

- Grep -c: count the lines where strings are matched with -c option

```
nikhilyadav@fedora:~$ cat >jungle.txt
Nikhil
Nikhil
Nikhil
Nikhil
^C
nikhilyadav@fedora:~$ grep -c "Nikhil" jungle.txt
4
```

REJEX COMMANDS

- ‘ ^ ’ :matches char at the beginning of the line

```
nikhilyadav@fedora:~$ grep ^N jungle.txt
Nikhil
Nikhil
Nikhil
nikhilyadav@fedora:~$
```

- character $ ‘ ’: matches char at the end of the line:

```
nikhilyadav@fedora:~$ grep 'l$' jungle.txt
Nikhil
Nikhil
Nikhil
```

- “.” Matches any character.

```
nikhilyadav@fedora:~$ grep "." jungle.txt
Nikhil
Nikhil
Nikhil
nikhilyadav@fedora:~$
```

- Step-[a-z] matches any char between a and z.

```
nikhilyadav@fedora:~$ egrep "[a-zA-Z]" jungle.txt
Find the treasure
Find the treasure
find the treasure9
Find the treasure
Find the junk
Find the jun2
Find the treasure
WOLVES
```

F- grep commands

- Grep -f (main_file file_to_search): matches and show same pattern from more than one file.

```
nikhilyadav@fedora:~$ cat >jungle.txt
error
warninh
failed
^C
nikhilyadav@fedora:~$ cat > story.txt
error
failed
^C
nikhilyadav@fedora:~$ grep -f story.txt jungle.txt
error
failed
nikhilyadav@fedora:~$
```

- Grep -f (-v) :invert match ….show the lines that do not match the patterns

```
nikhilyadav@fedora:~$ grep -f story.txt -v jungle.txt
warninh
nikhilyadav@fedora:~$
```

- Grep -f (-i) : ignore case sensitive over the files

```
nikhilyadav@fedora:~$ cat >story.txt
Hello
Welcome
^C
nikhilyadav@fedora:~$ cat >jungle.txt
hello
welcome
^C
nikhilyadav@fedora:~$ grep -f story.txt -i jungle.txt
hello
welcome
nikhilyadav@fedora:~$
```

E-grep commands

- Grep -e “pattern1” -e “pattern2” file_name: this commands allows you to specify multiple patterns to search in the input file.

```
nikhilyadav@fedora:~$ grep -e "directory." -e "failed" jungle.txt
failed to show error in his new directory.
nikhilyadav@fedora:~$
```

- Grep -e (-o): show only the part of the line that matches

```
nikhilyadav@fedora:~$ grep-e "directory." -e "faile" -o jungle.txt
bash: grep-e: command not found...
grep -e "nikhilyadav@fedora:~$ grep -e "directory" -e "failed" -o jungle.txt
failed
directory
nikhilyadav@fedora:~$
```

5. Learning outcomes (What I have learned):

a) Learnt how to use the GREP command to search for specific text patterns in files.
b) Learnt difference between GREP, EGREP, and FGREP, and know when to use each.
c) Learnt how to apply regular expressions with GREP to perform more advanced and flexible text searches.